

IP Address Practice Sheet - Answer Sheet

IP addressing | IPv4 and IPv6 | Subnetting | Classful and classless addressing

1. Classful and classless IPv4 addressing / subnetting

Supplementary practice question from the supplied practice sheet

- (a) Class B.
- (b) 255.255.0.0.
- (c) 255.255.255.192.
- (d) 192.168.10.64.
- (e) 192.168.10.127.

Working note: /26 gives blocks of 64 in the final octet: 0-63, 64-127, 128-191, 192-255. The host address .70 lies in the 64-127 subnet.

2. IP address types

Source: 9618/12 M/J/2021 Question 5(d)

Type	Most appropriate description
Public IP address	It is visible to any device on the internet.
Private IP address	It is only visible to devices within the Local Area Network (LAN).
Dynamic IP address	A new one is reallocated each time a device connects to the internet.
Static IP address	It does not change each time a device connects to the internet.

3. Subnetting benefits, subnet masks, network ID and host ID

Source: 9618/13 O/N/2022 Question 7(c)

(i) Any two accepted benefits include:

- improves security
- reduces congestion
- allows extension of the network / more devices to be attached
- aids day-to-day management
- improves performance

(ii) Network ID = 10; Host ID = 4.

4. IPv4 and IPv6 address structure; benefits of subnetting

Source: 9618/11 M/J/2023 Question 4(d)

(i) IPv4: 4 groups; 8 bits per group; groups separated by full stops.

IPv6: 8 groups; 16 bits per group; consecutive zero-only groups can be replaced by a double colon (::).

(ii) Two benefits, with expansion, can be chosen from:

- Reduce network traffic / improve network speed - data can remain within its subnet and travel less far.
- Improve network security - not all devices can access all areas of the network.

- Make maintenance easier - one subnetwork can be taken down or changed while the rest of the network continues.

5. Network ID and host ID in a subnetwork

Source: 9618/13 M/J/2023 Question 2(e)

- Network ID: identifies the subnetwork. Devices in the same subnetwork share the same network ID; different subnetworks have different network IDs.
- Host ID: uniquely identifies a device within its subnetwork. Devices in the same subnetwork have different host IDs.

6. IPv6, subnetting reasons and types of IP address

Source: 9618/12 O/N/2023 Question 7(a), 7(b)(iii), 7(d)

(a) 192.168.3.2 is not IPv6 because, for example:

- it has only four groups of digits rather than eight
- it uses dotted notation rather than colons
- it is a 32-bit / 4-byte form, whereas IPv6 is 128 bits / 16 bytes

(b)(iii) Two reasons for subnetting, each with a matching expansion, may include:

- Improve security - devices need not receive unintended data; a compromised device need not expose the whole network; access can be restricted between segments.
- Make network management easier - faults can be isolated more efficiently.
- Make the network easier to expand / improve control of network growth - a greater range of IP addresses can be made available.
- Improve performance / reduce congestion - communications are localised and data within the same subnet stays within that subnet.

(d) Complete table:

Type of IP address	Description
Public IP address	Assigned to a device to allow direct access on the internet.
Static IP address	Fixed; does not change each time a device re-joins a LAN / network.
Private IP address	Used for internal LAN communication only.
Dynamic IP address	May be refreshed / changed each time a device re-joins a LAN / network.

7. Public IP addresses and IPv4/IPv6 differences

Source: 9618/11 M/J/2024 Question 8(b)

(i) The router has a public IP address so that it is visible to and accessible by other devices on the internet.

(ii) Any two differences other than full stops/colons:

- IPv4 has 4 groups of digits; IPv6 has 8 groups.
- IPv4 is usually written in denary; IPv6 is usually written in hexadecimal.
- IPv4 groups range from 0 to 255; IPv6 groups range from 0 to FFFF.
- IPv4 is 32 bits; IPv6 is 128 bits.

8. Static private IP addresses

Source: 9618/13 M/J/2024 Question 5(d)

Static means the IP address for the device does not change. Private means it can only be accessed / seen / used within the LAN.

9. Purpose of public/private IP addresses and subnetting

Source: 9618/13 O/N/2024 Question 9(d)

(i) Public IP address: makes the router visible to the internet / external network / WAN.

Private IP address: identifies the router to computers within the LAN.

(ii) Accepted subnetting purposes include:

- Divides the network into smaller networks.
- Reduces traffic in some parts of the network / reduces congestion because traffic travels only through the parts necessary.
- Hides the complexity of the network.
- Allows easier maintenance of the network.

10. Complete the IPv4/IPv6 and IP-address-type description

Source: 9618/12 M/J/2025 Question 6(c)

Blank	Answer
1	full stops
2	hexadecimal
3	zeros
4	128
5	dynamic
6	private

11. Static/public IP addresses and IP-address validity

Source: 9618/11 O/N/2025 Question 7(a-b)

(a) Static: an IP address that does not change each time a device connects to the network.

Public: an IP address assigned to a device so that the device is visible on the internet.

(b) Invalid IP address.

- It is not IPv4 because an IPv4 denary group cannot exceed 255; the address uses 256.
- It is not IPv6 because it uses full stops rather than colons and does not have enough groups.

12. Static private IP addresses and benefits of subnetting

Source: 9618/11 M/J/2026 Question 7(c)

(i) A static private IP address does not change and is only visible / accessible within the network / LAN.

(ii) Any three sensible benefits, for example:

- Splits a larger network into smaller, more manageable parts.
- Can increase network efficiency.
- Reduces congestion / can reduce collisions.

- Improves security.
- Makes faults easier to isolate.
- Makes the network easier to scale.

13. Router tasks involving IP addresses

Source: 9618/11 M/J/2021 Question 4(c)(i) and 9618/13 M/J/2021 Question 4(c)(i) - identical item

Task	Performed by router?
Receives packets from devices	Yes
Finds the IP address of a Uniform Resource Locator (URL)	No
Directs each packet to all devices attached to it	No
Stores the IP and/or MAC address of all devices attached to it	Yes